E.G.S. PILLAY ENGINEERING COLLEGE, *(Autonomous)*
NAGAPATTINAM – 611 002.
(Affiliated to Anna University, Chennai | Accredited by NAAC with 'A++' Grade
Accredited by NBA | Approved by AICTE, New Delhi)

REGULATIONS - R2023
B.E. / B.Tech. – SIXTH SEMESTER CURRICULUM

Sixth Semester										
S.No	COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX. MARKS		
								CA	ES	TOTAL
Theory Courses										
1	2302ME601	Heat and Mass Transfer	PCC	2	1	0	3	40	60	100
2	2302ME602	Design of Transmission Systems	PCC	3	1	0	4	40	60	100
3	2302ME603	Dynamics of Machines	PCC	2	1	0	3	40	60	100
4	2302ME604	Mechatronics	PCC	3	0	0	3	40	60	100
5		Professional Elective - III	PEC	3	0	0	3	40	60	100
6		Open Elective – II	OEC	3	0	0	3	40	60	100
7		Mandatory Course -II	MC	3	0	0	0	100	-	100
Laboratory Courses										
8	2302ME651	Dynamics Laboratory	PCC	0	0	2	1	60	40	100
9	2302ME652	Mechatronics Laboratory	PCC	0	0	2	1	60	40	100
10	2302ME653	Heat and Mass Transfer Laboratory	PCC	0	0	2	1	60	40	100
11	2304GE601	Professional Development course -IV	EEC	0	0	2	1	100	-	100
12	2301LS601	Life skill course VI	MC	0	0	0	0	100	-	100
			TOTAL	**19**	**3**	**8**	**23**	**720**	**480**	**1200**

2302ME601	HEAT AND MASS TRANSFER	L	T	P	C
		2	1	0	3

PREREQUISITE

1. Engineering Physics
2. Engineering Mathematics
3. Thermodynamics
4. Fluid Mechanics

COURSE OBJECTIVES:

1. To Learn the principal mechanism of heat transfer under steady state and transient conditions.
2. To learn the fundamental concept and principles in convective heat transfer.
3. To learn the theory of phase change heat transfer and design of heat exchangers.
4. To study the fundamental concept and principles in radiation heat transfer.
5. To develop the basic concept and diffusion, convective di mass transfer.

MODULE I	**CONDUCTION**	**9 Hours**

General Differential equation of Heat Conduction– Cartesian and Polar Coordinates – One Dimensional Steady State Heat Conduction — plane and Composite Systems – Conduction with Internal Heat Generation – Extended Surfaces – Unsteady Heat Conduction – Lumped Analysis – Semi Infinite and Infinite Solids –Use of Heisler's charts-overview of Computational fluid dynamics and its application in steady state conduction problem- Analysis of heat transfer in sustainable materials and energy-efficient insulation systems

MODULE II	**CONVECTION**	**9 Hours**

Free and Forced Convection – Hydrodynamic and Thermal Boundary Layer. Free and Forced Convection during external flow over Plates and Cylinders and Internal flow through tubes – key features of ANSYS Fluent for convection flow analysis-Optimization of convection processes in renewable energy systems.

MODULE III	**PHASE CHANGE HEAT TRANSFER AND HEAT EXCHANGERS**	**9 Hours**

Nusselt"s theory of condensation – Regimes of Pool boiling and Flow boiling. Correlations in boiling and condensation. Heat Exchanger Types – Overall Heat Transfer Coefficient – Fouling Factors – Analysis – LMTD method – NTU method- usage of Aspen Plus for Heat exchanger design and analysis- Development of energy-efficient heat exchangers and their application in sustainable industries.

MODULE IV	**RADIATION**	**9 Hours**

Black Body Radiation – Grey body radiation – Shape Factor – Electrical Analogy – Radiation Shields. Radiation through gases- Radiative heat transfer analysis using Monte Carlo simulation methods- Applications of radiation heat transfer in solar energy and greenhouse gas modeling.

MODULE V	**MASS TRANSFER**	**9 Hours**

Basic Concepts – Diffusion Mass Transfer – Fick"s Law of Diffusion – Steady state Molecular Diffusion– Convective Mass Transfer – Momentum, Heat and Mass Transfer Analogy –Convective Mass Transfer Correlations- key features of COMSOL Multiphysics for mass transfer analysis- Study of mass transfer processes in environmental systems, such as pollutant dispersion and carbon capture technologies.

TOTAL: 45 HOURS

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Analyze the rate of heat transfer under steady-state and unsteady-state heat conduction conditions.
CO2:	Apply convection principles and heat transfer correlations to internal and external flows under free and forced convection condition.
CO3:	Examine heat transfer in boiling, condensation, and heat exchangers using correlations and LMTD/NTU methods for sustainable applications.
CO4:	Evaluate radiative heat exchange between surfaces and gases using radiation laws, shape factors,

	and shielding concepts.
CO5:	Assess mass transfer by diffusion and convection in engineering and environmental applications using appropriate correlations and analogies.

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	1	1	1	1	-	-	-	3
CO2	3	3	3	1	1	1	1	-	-	-	3
CO3	3	3	3	1	1	1	1	-	-	-	3
CO4	3	3	3	1	1	1	1	-	-	-	3
CO5	3	3	3	1	1	1	1	-	-	-	3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	3	-	-
CO2	3	-	-
CO3	3	-	-
CO4	3	-	-
CO5	3	-	-

REFERENCES:

1. R.C. Sachdeva, “Fundamentals of Engineering Heat & Mass transfer”, New Age International Publishers, 2009
2. Yunus A. Cengel, “Heat Transfer A Practical Approach” – Tata McGraw Hill, 5thEdition – 2013
3. Frank P. Incropera and David P. Dewitt, “Fundamentals of Heat and Mass Transfer”, John Wiley & Sons, 7th Edition, 2014.
4. Holman, J.P., “Heat and Mass Transfer”, Tata McGraw Hill, 2010
5. Kothandaraman, C.P., “Fundamentals of Heat and Mass Transfer”, New Age International, New Delhi, 2012
6. Ozisik, M.N., “Heat Transfer”, McGraw Hill Book Co., 1994. 5. S.P. Venkateshan, “Heat Transfer”, Ane Books, New Delhi, 2014

2302ME602	DESIGN OF TRANSMISSION SYSTEMS	L	T	P	C
		3	1	0	4

PREREQUISITE

1. Engineering Mechanics
2. Strength of Materials
3. Theory of Machines
4. Design of Machine Elements

COURSE OBJECTIVES:

1. To gain knowledge on the principles and procedure for the design of Mechanical power Transmission components.
2. To understand the standard procedure available for Design of Transmission of Mechanical elements spur gears and parallel axis helical gears.
3. To learn the design bevel, worm and cross helical gears of Transmission system.
4. To learn the concepts of design multi and variable speed gear box for machine tool applications.
5. To learn the concepts of design to cams, brakes and clutches

MODULE I	DESIGN OF FLEXIBLE ELEMENTS	9 + 3 Hours

Design of Flat belts and pulleys – Selection of V belts and pulleys – Selection of hoisting wire ropes and pulleys – Design of Transmission chains and Sprockets.

MODULE II	SPUR GEARS AND PARALLEL AXIS HELICAL GEARS	9 + 3 Hours

Speed ratios and number of teeth-Force analysis -Tooth stresses – Dynamic effects – Fatigue strength – Factor of safety – Gear materials – Design of straight tooth spur & helical gears based on strength and wear considerations – Pressure angle in the normal and transverse plane-Equivalent number of teeth-forces for helical gears.

MODULE III	BEVEL, WORM AND CROSS HELICAL GEARS	9 + 3 Hours

Straight bevel gear: Tooth terminology, tooth forces and stresses, equivalent number of teeth. Estimating the dimensions of pair of straight bevel gears. Worm Gear: Merits and demerits terminology. Thermal capacity, materials-forces and stresses, efficiency, estimating the size of the worm gear pair. Cross helical: Terminology-helix angles-Estimating the size of the pair of cross helical gears.

MODULE IV	GEAR BOXES	9 + 3 Hours

Geometric progression – Standard step ratio – Ray diagram, kinematics layout -Design of sliding mesh gear box – Design of multi speed gear box for machine tool applications – Constant mesh gear box – Speed reducer unit. – Variable speed gear box, Fluid Couplings, Torque Converters for automotive applications.

MODULE V	CAMS, CLUTCHES AND BRAKES	9 + 3 Hours

Cam Design: Types-pressure angle and under cutting base circle determination-forces and surface stresses. Design of plate clutches –axial clutches-cone clutches-internal expanding rim clutches-Electromagnetic clutches. Band and Block brakes – external shoe brakes – Internal expanding shoe brake.

TOTAL: 60 HOURS

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Apply the concepts of design to belts, chains and rope drives.
CO2:	Apply the concepts of design to spur, helical gears.
CO3:	Apply the concepts of design to worm and bevel gears.
CO4:	Apply the concepts of design to gear boxes.
CO5:	Apply the concepts of design to cams, brakes and clutches

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	3	1	-	-	-	-	1	-	1
CO2	3	2	3	1	-	-	-	-	1	-	1
CO3	3	2	3	1	-	-	-	-	1	-	1
CO4	3	2	3	1	-	-	-	-	1	-	1
CO5	3	2	3	1	-	-	-	-	1	-	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	-	3
CO2	-	-	3
CO3	-	-	3
CO4	-	-	3
CO5	-	-	3

REFERENCES:

1. Bhandari V, "Design of Machine Elements", 4th Edition, Tata McGraw-Hill Book Co, 2016.
2. Joseph Shigley, Charles Mischke, Richard Budynas and Keith Nisbett "Mechanical Engineering Design", 8th Edition, Tata McGraw-Hill, 2008.
3. Merhyle F. Spotts, Terry E. Shoup and Lee E. Hornberger, "Design of Machine Elements" 8th Edition, Printice Hall, 2003.
4. Orthwein W, "Machine Component Design", Jaico Publishing Co, 2003.
5. Prabhu. T.J., "Design of Transmission Elements", Mani Offset, Chennai, 2000.
6. Robert C. Juvinall and Kurt M. Marshek, "Fundamentals of Machine Design", 4th Edition, Wiley,2005
7. Sundararajamoorthy T. V, Shanmugam .N, "Machine Design", Anuradha Publications, Chennai, 2003.

2302ME603	DYNAMICS OF MACHINES	L	T	P	C
		2	1	0	3

PREREQUISITE

1. Engineering Mathematics
2. Engineering Mechanics
3. Strength of Materials
4. Kinematics of Machines

COURSE OBJECTIVES:

1. To understand the force-motion relationship in components subjected to external forces and analysis of standard mechanisms.
2. To understand the undesirable effects of unbalances resulting from prescribed motions in mechanism.
3. To understand the principles in mechanisms used for speed control and stability control.
4. To understand the effect of Dynamics of undesirable vibrations.
5. To understand the basic principles of Transverse and Torsional vibrations

MODULE I	DYNAMIC FORCE ANALYSIS OF MECHANISMS	9 Hours

Principle of superposition, Condition for dynamic analysis, Dynamic analysis of four bar & slider crank mechanism - Engine force analysis. Turning moment diagram for steam & IC Engine. Energy stored in flywheel, Dimension of flywheel rim, Flywheel in punching press.

MODULE II	BALANCING	9 Hours

Introduction - Static balancing and dynamic balancing, Balancing of Rotating mass several masses in same and different plane. Balancing of reciprocating mass Swaying couple, Tractive force, Hammer Blow. Balancing of coupled locomotives.

MODULE III	GOVERNOR AND GYROSCOPE	9 Hours

Governor Terminology, working principle, Types - Watt, Porter and Proell governor, Characteristics of Governor-sensitiveness, Hunting, Ichoronisn, Stability. Gyroscope- Gyroscopic effect, gyroscopic couple, gyroscopic effect on aero planes and naval ships.

MODULE IV	FUNDAMENTAL OF VIBRATION	9 Hours

Introduction-Terminology, Classification, elements of vibration, free undamped vibration, Free Damped vibration (Viscus Damping) - Damping ratio and logarithmic decrement. Force damped vibration - Magnification factor. Vibration isolation and transmissibility.

MODULE V	TRANSVERSE AND TORSIONAL VIBRATION	9 Hours

Transverse vibration of shafts and beams, Shaft carrying several loads, whirling of shafts. Torsional vibration- effect of inertia on torsional vibration-Torsionally equivalent Shaft, single rotor, two rotors and three rotors system.

TOTAL: 45 HOURS

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Calculate the dynamic force analysis of different mechanisms.
CO2:	Calculate the balancing masses and their locations of reciprocating and rotating masses.
CO3:	Calculate the speed and lift of the governor and estimate the gyroscopic effect on ships and airplanes.
CO4:	Compute the frequency of free vibration, forced vibration and damping coefficient.
CO5:	Compute the effect of inertia force on transverse and torsional vibrations for single, two and three rotor systems.

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	-	-	1	-	-	-	2

CO2	3	3	2	2	-	-	1	-	-	-	2
CO3	3	3	2	2	-	-	1	-	-	-	2
CO4	3	3	2	2	-	-	1	-	-	-	2
CO5	3	3	2	2	-	-	1	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	-	3
CO2	-	-	3
CO3	-	-	3
CO4	-	-	3
CO5	-	-	3

REFERENCES:

1. Uicker, J.J., Pennock G.R and Shigley, J.E., “Theory of Machines and Mechanisms” ,3rd Edition, Oxford University Press, 2009.
2. Rattan, S.S, “Theory of Machines”, 3rd Edition, Tata McGraw-Hill, 2009
3. Thomas Bevan, "Theory of Machines", 3rd Edition, CBS Publishers and Distributors, 2005.
4. Cleghorn. W. L, “Mechanisms of Machines”, Oxford University Press, 2005
5. Benson H. Tongue, “Principles of Vibrations”, Oxford University Press, 2nd Edition, 2007
6. http://nptel.ac.in/12106166.

2302ME604	MECHATRONICS	L	T	P	C
		3	0	0	3

Prerequisite

1. Basic Electrical and Electronics Engineering
2. Hydraulics and Pneumatics

Course Objectives:

1. To make students get acquainted with the sensors and the actuators, which are commonly used in mechatronics systems.
2. To provide insight into the signal conditioning circuits, and also to develop competency in PLC programming and control
3. To make students familiarize with the fundamentals of IoT and Embedded systems.
4. To impart knowledge about the Arduino and the Raspberry Pi.
5. To inculcate skills in the design and development of mechatronics and IoT based systems.

MODULE I	**SENSORS**	**9 Hours**

Components of mechatronics system, Sensor – terminology and Mathematical equation – Potentiometer, Linear Variable differential transformer, strain gauge, Piezoelectric sensor, Optical encoder, Hall effect sensor, Thermistor, Thermo-couple, Light sensor.

MODULE II	**ACTUATORS**	**9 Hours**

Terminology, mathematical equation of Mechanical Actuation system – cam, gear, belt & chain, Ball screw, Mechanical aspects of motor selection. Pneumatic & hydraulic Actuation system. Electrical actuation system - relay & solenoid, working & control of Brush & brushless DC motor, working & control of Stepper & servo motor.

MODULE III	**FEEDBACK CONTROL**	**9 Hours**

Transfer Function, Mathematical Modeling of Mechanical & Electrical system, Electrical analogy, Electro-mechanical system, First order system, second order system, Proportional control, derivative control, Integral control, PID control, Controller tuning, Concept of stability.

MODULE IV	**MICROCONTROLLER**	**9 Hours**

Architecture of 8051- I/O Pins, Ports and Circuits, memory, counter, Timer, Interrupt, Instruction set- Moving data, Logical, arithmetic operation, Jump & call instruction, LCD & Keyboard Interfacing. Examples - Windscreen wiper motion, Car engine management.

MODULE V	**PROGRAMMABLE LOGIC CONTROLLER**	**9 Hours**

Basic Structure – Input / Output Processing – Programming – Mnemonics – Timers, Internal relays and counters – Shift Registers – Master and Jump Controls – Data Handling – Analogue Input / Output – Selection of PLC. Examples -Pick and place robot. Car park barrier system.

TOTAL: 45 HOURS

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Explain Select suitable sensors and actuators to develop mechatronics systems.
CO2:	Discuss Devise proper signal conditioning circuit for mechatronics systems, and also able to implement PLC as a controller for an automated system.
CO3:	Elucidate the fundamentals of IoT and Embedded Systems.
CO4:	Discuss Control I/O devices through Arduino and Raspberry Pi.
CO5:	Design and develop an apt mechatronics/IoT based system for the given real-time application.

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	1	-	-	-	-	-	
CO2	3	3	3	1	2	-	-	-	1	-	2
CO3	3	1	2	1	2	-	2	-	-	-	
CO4	3	3	3	3	3	-	-	-	3	-	3
CO5	3	3	3	3	3	-	2	-	3	-	3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	-	3
CO2	-	-	3
CO3	-	-	3
CO4	-	-	3
CO5	-	-	3

REFERENCES:

1. Bradley D.A., Burd N.C., Dawson D., Loader A.J., “Mechatronics: Electronics in Products and Processes”, Routledge, 2017.
2. Sami S.H and Kisheen Rao G “The Internet of Mechanical Things: The IoT Framework for Mechanical Engineers”, CRC Press, 2022.
3. John Billingsley, “Essentials of Mechatronics”, Wiley, 2006
4. David H., Gonzalo S., Patrick G.,Rob B. and Jerome H.,“IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things”, Pearson Education, 2018.
5. Nitin G and Sharad S, “Internet of Things: Robotic and Drone Technology”, CRC Press, 2022
6. Newton C. Braga, “Mechatronics for The Evil Genius”, McGrawHill, 2005.
7. Bell C., “Beginning Sensor Networks with Arduino and Raspberry Pi”, A press, 2013

2302ME651	DYNAMICS LABORATORY	L	T	P	C
		0	0	2	1

PREREQUISITE

1. Engineering Mathematics
2. Engineering Mechanics
3. Kinematics of Machines

COURSE OBJECTIVES:

1. To supplement the principles learnt in kinematics and Dynamics of Machinery.
2. To understand how certain measuring devices are used for dynamic testing.
3. To study the working principle of governor and gyroscope.
4. To learn the concept of free and forced vibration.
5. To learn the concept of transverse and torsion vibration.

LIST OF EXPERIMENTS

1. Determination of mass moment of inertia of axisymmetric bodies using turn table apparatus
2. Determine the characteristics and effort of Watt, Porter Proell and Hartnell Governors
3. Exercise on Balancing of reciprocating masses.
4. Exercise on Balancing of four rotating masses placed on different plane.
5. Analyze the gyroscopic effect using Gyroscope and verify its laws.
6. Determination of critical speed of shaft with concentrated loads by Whirling of shaft & vibration table apparatus.
7. Determine the moment of inertia of object by Bifilar suspension, Trifilar & method of oscillation.
8. Kinematic analysis of cam model, Epicycle gear train and differential model.
9. Determination of natural frequency of single degree of freedom system & two rotor system
10. Determine the frequency of forced vibration using Cantilever beam.
11. Determination of natural frequency of Torque measurement system.

TOTAL: 30 HOURS

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Measure the mass moment of inertia of axisymmetric objects using Turn table apparatus, bi-filar suspension and compound pendulum.
CO2:	Experiment with vibrations and balancing of rotating and reciprocating masses in dynamic balancing machine.
CO3:	Make use of Watt, Porter, Proell and Hartnell governors for determination of range sensitivity.
CO4:	Do experimentation of the critical speed of shaft under the given load conditions.
CO5:	Perform the torsional natural frequency of single and double rotor systems.
CO6:	Make use of whirling of shaft for determination of critical speed of a shaft with concentrated loads.
CO7:	Measure the transmissibility ratio using vibrating table.

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	1	-	-	3	-	3	-	2
CO2	3	2	2	1	-	-	3	-	3	-	2
CO3	3	2	2	1	-	-	3	-	3	-	2
CO4	3	2	2	1	-	-	3	-	3	-	2

CO5	3	2	2	1	-	-	3	-	3	-	2
CO6	3	2	2	1	-	-	3	-	3	-	2
CO7	3	2	2	1	-	-	3	-	3	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	3	3
CO2	-	3	3
CO3	-	3	3
CO4	-	3	3
CO5	-	3	3
CO6	-	3	3
CO7	-	3	3

2302ME652	MECHATRONICS LABORATORY	L	T	P	C
		0	0	2	1

PREREQUISITE

1. Basic Electrical and Electronics Engineering.
2. Hydraulics and Pneumatics.

COURSE OBJECTIVES:

1. To study the concept of mechatronics to design, modelling and analysis of basic electrical hydraulic systems.
2. To provide the hands on-training in the control of linear and rotary actuators.
3. To study the concepts and fundamentals of IoT, sensors, actuators and IoT boards

LIST OF EXPERIMENTS

1. Measurement of Linear/Angular of Position, Direction and Speed using Transducers.
2. Measurement of Pressure, Temperature and Force using Transducers.
3. Speed and Direction control of DC Servomotor, AC Servomotor and Induction motors.
4. Addition, Subtraction and Multiplication Programming in 8051.
5. Programming and Interfacing of Stepper motor and DC motor using 8051/PLC.
6. Programming and Interfacing of Traffic Light Interface using 8051.
7. Sequencing of Hydraulic and Pneumatic circuits.
8. Sequencing of Hydraulic, Pneumatic and Electro-pneumatic circuits using Software.
9. Electro-pneumatic/hydraulic control using PLC.

TOTAL: 30 HOURS

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Demonstrate the functioning of mechatronics systems with various pneumatic, hydraulic and electrical systems.
CO2:	Demonstrate the microcontroller and PLC as controllers in automation systems by executing proper interfacing of I/O devices and programming
CO3:	Demonstrate the sensing and actuation of mechatronics elements using IoT.

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	1	1	1	3	-	-	-	3	-	3
CO2	1	1	1	1	3	-	-	-	3	-	3
CO3	1	1	3	3	3	-	-	-	3	-	3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	1	3
CO2	1	1	3
CO3	1	1	3

2302ME653	HEAT AND MASS TRANSFER LABORATORY	L	T	P	C
		0	0	2	1

PREREQUISITE

1. Engineering Mathematics
2. Engineering Physics
3. Thermodynamics
4. Thermal Engineering
5. Fluid Mechanics

COURSE OBJECTIVES:

1. To gain experimental knowledge of Predicting the thermal conductivity of solids and liquids.
2. To gain experimental knowledge of Estimating the heat transfer coefficient values of various fluids.
3. To gain experimental knowledge of Testing the performance of tubes in tube heat exchangers.

LIST OF EXPERIMENTS

1. Determination of thermal conductivity of insulating powder.
2. Determination of thermal conductivity of guarded hot plate.
3. Determination of thermal conductivity of materials in lagged pipe.
4. Determination of heat transfer co-efficient through composite wall.
5. Determination of heat transfer co-efficient by natural convection.
6. Determination of heat transfer co-efficient by forced convection.
7. Determination of heat transfer co-efficient in a parallel and counter flow heat exchanger.
8. Determination of heat transfer co-efficient and effectiveness from Pin-Fin by natural convection.
9. Determination of heat transfer co-efficient and effectiveness from Pin-Fin by forced convection.
10. Determination of Stefan-Boltzmann constant.
11. Determination of emissivity using emissivity apparatus.
12. Determination of performance in a fluidized bed cooling tower.

TOTAL: 30 HOURS

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Conduct experiment on Predict the thermal conductivity of solids and liquids.
CO2:	Conduct experiment on Estimate the heat transfer coefficient values of various fluids
CO3:	Conduct experiment on Test the performance of tubes in tube heat exchangers.

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	1	3	2	-	-	-	-	1	-	1
CO2	1	1	3	2	-	-	-	-	1	-	1
CO3	1	1	3	2	-	-	-	-	1	-	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	3	-	-
CO2	3	-	-
CO3	3	-	-